



# PERCONA

Databases run better with Percona

ProxySQL no MySQL: Apenas  
um load balancer?





# Agenda

# Agenda

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A large, stylized, light orange logo is positioned on the left side of the image. It consists of a thick, curved line that forms a shape resembling a stylized 'A' or a mountain peak, with a circular element integrated into its upper right portion.

Sobre mim

# Roberto Garcia de Bem



Engenheiro de Suporte  
na Percona desde 2022

DBA durante quase 4  
anos na Performance DB

Trabalhando com MySQL  
mais de 5 anos

Sistemas de Informação  
- Universidade do Sul de  
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O que é o ProxySQL?

# O que é o ProxySQL?

A High Performance MySQL Proxy: ProxySQL is an open source high performance, high availability, database protocol aware proxy for MySQL

Um MySQL Proxy de Alta Performance: ProxySQL é um proxy de código aberto de alto desempenho e alta disponibilidade com reconhecimento de protocolo de banco de dados para MySQL. (Tradução Livre)

Criado e mantido por: René Canhaõ

<https://github.com/sysown/proxysql>

<https://proxysql.com/>





# ProxySQL Instalação

- Source code / RPM / DEB / Percona repo (yum / apt)

<https://proxysql.com/documentation/installing-proxysql/>

<https://www.percona.com/download-proxysql>

<https://docs.percona.com/percona-software-repositories/index.html>



# Features ProxySQL



# Features ProxySQL

Load Balance

# Load Balance

## Divisão da Leitura e escrita

```
[root@roberto-garciadebem-node3 ~]# mysql -uroot -p'verysecretpassword1^' -h127.0.0.1 -P6033 -e'INSERT INTO test.t1 VALUES (null,@@hostname)'
```

```
[root@roberto-garciadebem-node3 ~]# mysql -uroot -p'verysecretpassword1^' -h127.0.0.1 -P6033 -e'SELECT @@HOSTNAME,hostname as node FROM test.t1'
```

| @@HOSTNAME                | node                        |
|---------------------------|-----------------------------|
| roberto-garciadebem-node1 | roberto-garciadebem-default |

```
[root@roberto-garciadebem-node3 ~]# mysql -uroot -p'verysecretpassword1^' -h127.0.0.1 -P6033 -e'SELECT @@HOSTNAME,hostname as node FROM test.t1'
```

| @@HOSTNAME                | node                        |
|---------------------------|-----------------------------|
| roberto-garciadebem-node2 | roberto-garciadebem-default |

# Load balance – principais tabelas

- mysql\_replication\_hostgroups/mysql\_group\_replication\_hostgroups/  
mysql\_galera\_hostgroups

```
ProxySQL > SELECT * FROM mysql_replication_hostgroups;
```

| writer_hostgroup | reader_hostgroup | check_type | comment |
|------------------|------------------|------------|---------|
| 1                | 2                | read_only  |         |

```
1 row in set (0.00 sec)
```

# Load balance – principais tabelas

- `mysql_servers`

```
ProxySQL > select hostgroup_id,hostname,status,weight from mysql_servers;
```

| hostgroup_id | hostname     | status | weight |
|--------------|--------------|--------|--------|
| 1            | 10.124.33.61 | ONLINE | 1      |
| 2            | 10.124.33.69 | ONLINE | 1      |
| 2            | 10.124.33.22 | ONLINE | 1      |

```
3 rows in set (0.00 sec)
```

# Load balance – principais tabelas

- mysql\_query\_rules

```
ProxySQL > SELECT rule_id,active,match_pattern,destination_hostgroup,apply FROM mysql_query_rules;
```

| rule_id | active | match_pattern         | destination_hostgroup | apply |
|---------|--------|-----------------------|-----------------------|-------|
| 1       | 1      | ^SELECT .*            | 2                     | 0     |
| 2       | 1      | ^SELECT .* FOR UPDATE | 1                     | 0     |

2 rows in set (0.00 sec)

# Load balance – principais tabelas

- mysql\_user

```
ProxySQL > SELECT username,password,active,default_hostgroup FROM mysql_users;
```

| username | password             | active | default_hostgroup |
|----------|----------------------|--------|-------------------|
| root     | verysecretpassword1^ | 1      | 1                 |

```
1 row in set (0.00 sec)
```



# Load Balance

## Configuração dos Hostgroups/Servers

```
ProxySQL > select * from runtime_mysql_replication_hostgroups;
```

| writer_hostgroup | reader_hostgroup | check_type | comment |
|------------------|------------------|------------|---------|
| 1                | 2                | read_only  |         |

1 row in set (0.00 sec)

```
ProxySQL > select hostgroup_id,hostname from runtime_mysql_servers;
```

| hostgroup_id | hostname     |
|--------------|--------------|
| 1            | 10.124.33.61 |
| 2            | 10.124.33.22 |
| 2            | 10.124.33.69 |

3 rows in set (0.00 sec)

```
ProxySQL > select rule_id,active,match_pattern,destination_hostgroup from runtime_mysql_query_rules;
```

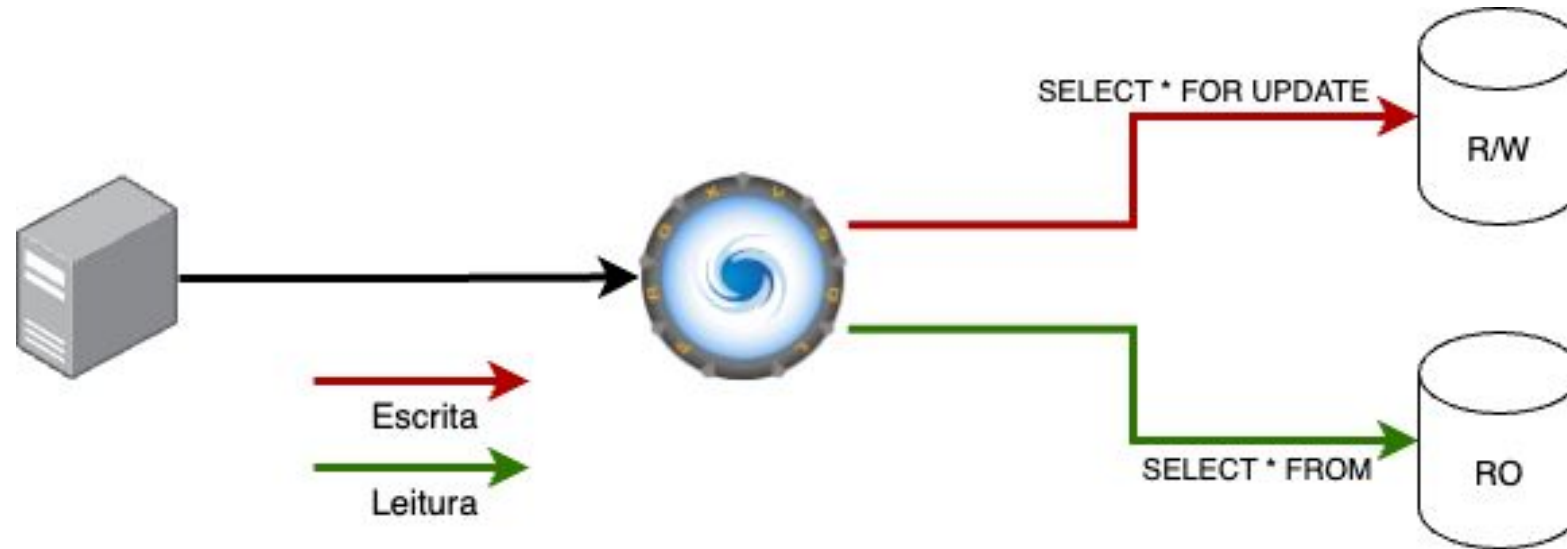
| rule_id | active | match_pattern         | destination_hostgroup |
|---------|--------|-----------------------|-----------------------|
| 1       | 1      | ^SELECT .*            | 2                     |
| 2       | 1      | ^SELECT .* FOR UPDATE | 1                     |

2 rows in set (0.00 sec)

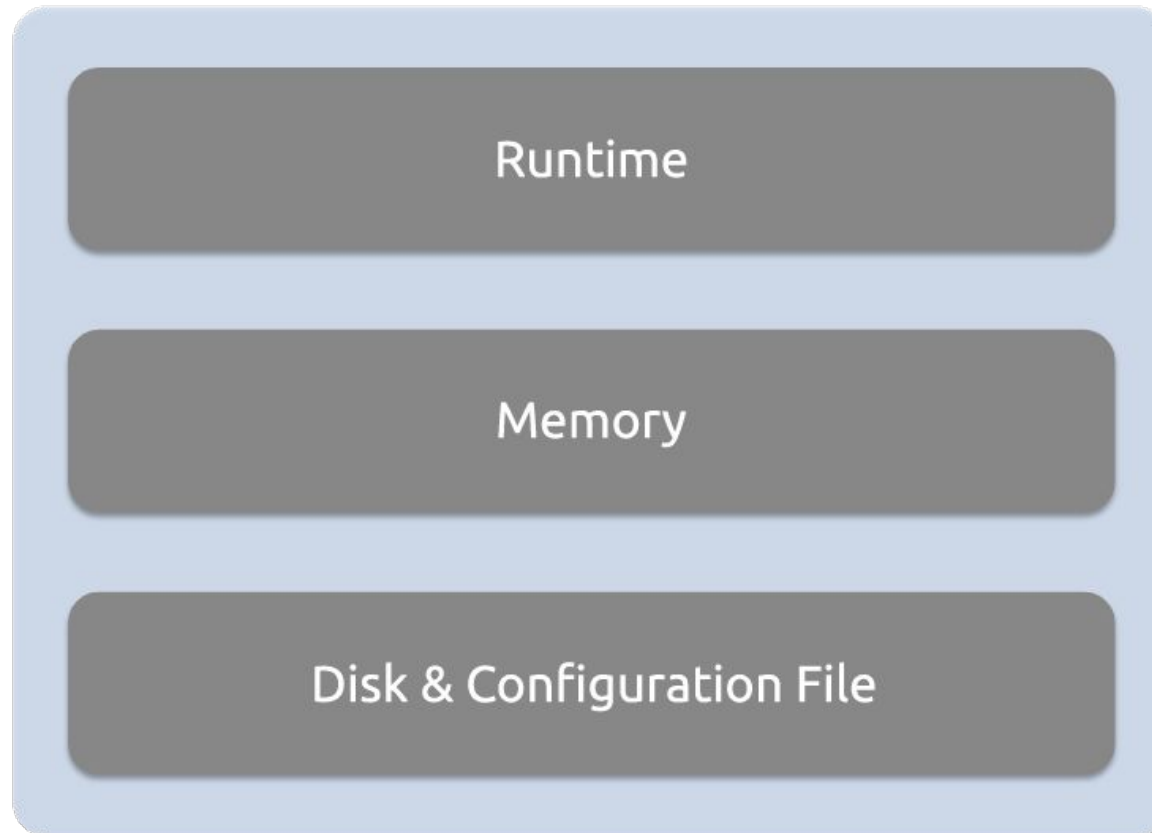
| NAME                        | STATE   | IPV4                 |
|-----------------------------|---------|----------------------|
| roberto-garciadebem-default | RUNNING | 10.124.33.61 (eth0)  |
| roberto-garciadebem-node1   | RUNNING | 10.124.33.22 (eth0)  |
| roberto-garciadebem-node2   | RUNNING | 10.124.33.69 (eth0)  |
| roberto-garciadebem-node3   | RUNNING | 10.124.33.232 (eth0) |

# Load Balance

Divisão da Leitura e escrita



# Camadas ProxySQL

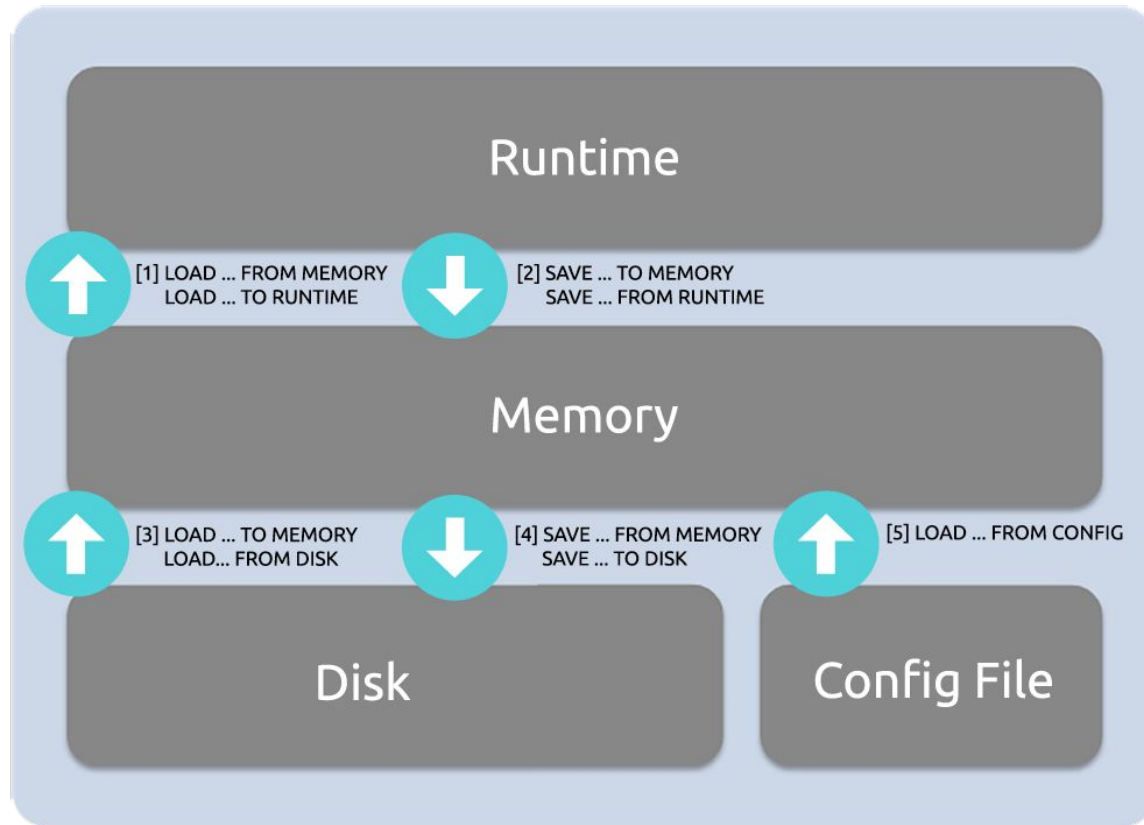


**Runtime:** Representa os valores em memória do ProxySQL. Tabelas iniciadas com runtime.

**Memory:** Representa o banco de dados em memória onde editamos configurações no ProxySQL.

**Disk:** Representa o banco de dados em disco (SQLite3).

# Alterando configurações entre camadas



Utilizamos os comandos LOAD e SAVE para alterar uma configuração entre camadas.

*Nota: mudanças são ativadas quando estão em runtime e mudanças não salvas em disco serão perdidas em caso de restart.*

# Alterando configurações entre camadas – Runtime

```
ProxySQL > INSERT INTO mysql_users (username,password,active,default_hostgroup) VALUES ('sandbox','sandbox',1,1);
Query OK, 1 row affected (0.00 sec)
```

```
ProxySQL > SELECT username,password,active,default_hostgroup FROM runtime_mysql_users;
```

| username | password                                  | active | default_hostgroup |
|----------|-------------------------------------------|--------|-------------------|
| root     | *A4EAAFACAC7E8241A4C7C903D8E679158895BD32 | 1      | 1                 |
| root     | *A4EAAFACAC7E8241A4C7C903D8E679158895BD32 | 1      | 1                 |

```
2 rows in set (0.00 sec)
```

```
ProxySQL > LOAD MYSQL USERS TO RUNTIME;
Query OK, 0 rows affected (0.00 sec)
```

```
ProxySQL > SELECT username,password,active,default_hostgroup FROM runtime_mysql_users;
```

| username | password                                  | active | default_hostgroup |
|----------|-------------------------------------------|--------|-------------------|
| root     | *A4EAAFACAC7E8241A4C7C903D8E679158895BD32 | 1      | 1                 |
| sandbox  | *BCA924B5D75F31FC6310837B9AFBB64022D1A0E9 | 1      | 1                 |
| root     | *A4EAAFACAC7E8241A4C7C903D8E679158895BD32 | 1      | 1                 |
| sandbox  | *BCA924B5D75F31FC6310837B9AFBB64022D1A0E9 | 1      | 1                 |

```
4 rows in set (0.00 sec)
```

# Alterando configurações entre camadas – Disk

```
[root@roberto-garciadebem-node3 ~]# systemctl restart proxysql
[root@roberto-garciadebem-node3 ~]# mysql -uadmin -padmin -h127.0.0.1 -P6032 --prompt='ProxySQL > '
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 1
Server version: 5.5.30 (ProxySQL Admin Module)
```

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
ProxySQL > SELECT username,password,active,default_hostgroup FROM mysql_users;
```

|          |                      |        |                   |
|----------|----------------------|--------|-------------------|
| username | password             | active | default_hostgroup |
| root     | verysecretpassword1^ | 1      | 1                 |

```
1 row in set (0.00 sec)
```

# Alterando configurações entre camadas – Disk

```
ProxySQL > INSERT INTO mysql_users (username,password,active,default_hostgroup) VALUES ('sandbox','sandbox',1,1);
Query OK, 1 row affected (0.00 sec)
```

```
ProxySQL > SAVE MYSQL USERS TO DISK;
Query OK, 0 rows affected (1.12 sec)
```

```
ProxySQL > EXIT
```

Bye

```
[root@roberto-garciadebem-node3 ~]# systemctl restart proxysql
```

```
[root@roberto-garciadebem-node3 ~]# mysql -uadmin -padmin -h127.0.0.1 -P6032 --prompt='ProxySQL > '
```

Welcome to the MariaDB monitor. Commands end with ; or \g.

Your MySQL connection id is 1

Server version: 5.5.30 (ProxySQL Admin Module)

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
ProxySQL > SELECT username,password,active,default_hostgroup FROM mysql_users;
```

| username | password             | active | default_hostgroup |
|----------|----------------------|--------|-------------------|
| root     | verysecretpassword1^ | 1      | 1                 |
| sandbox  | sandbox              | 1      | 1                 |

```
2 rows in set (0.00 sec)
```



# Alterando configurações entre camadas – Variáveis

Temos dois tipos de variáveis no ProxySQL:

- Admin
- MySQL

```
ProxySQL > SELECT * FROM global_variables LIMIT 115, 2;
```

| variable_name                         | variable_value |
|---------------------------------------|----------------|
| mysql-query_cache_stores_empty_result | true           |
| admin-stats_credentials               | stats:stats    |

```
2 rows in set (0.00 sec)
```



# Alterando configurações entre camadas – Variáveis

MySQL

```
ProxySQL > SELECT * FROM global_variables where variable_name like 'mysql-wait_timeout';
```

# Alterando configurações entre camadas – Variáveis

Admin

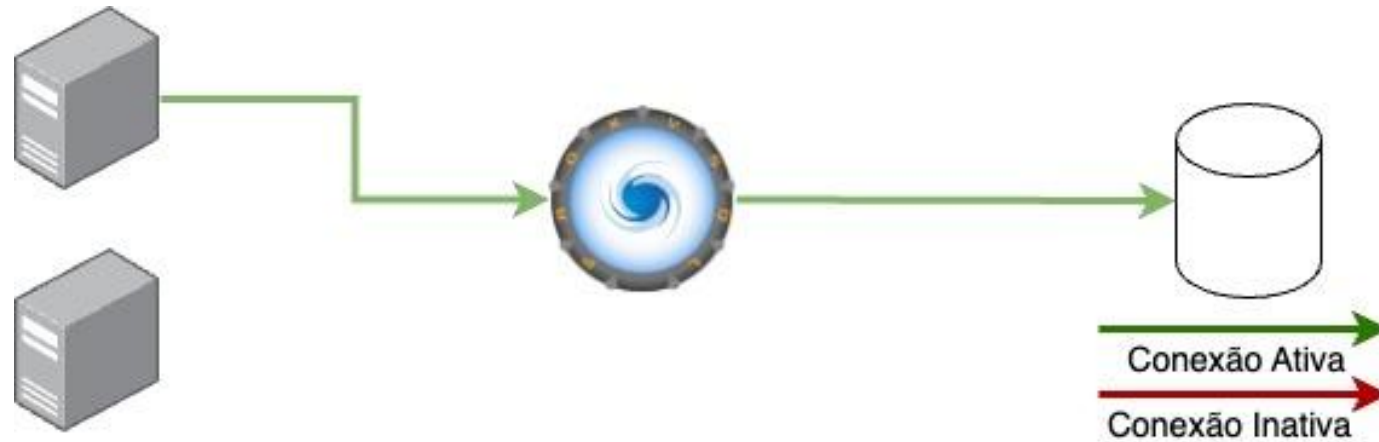
ProxySQL > |



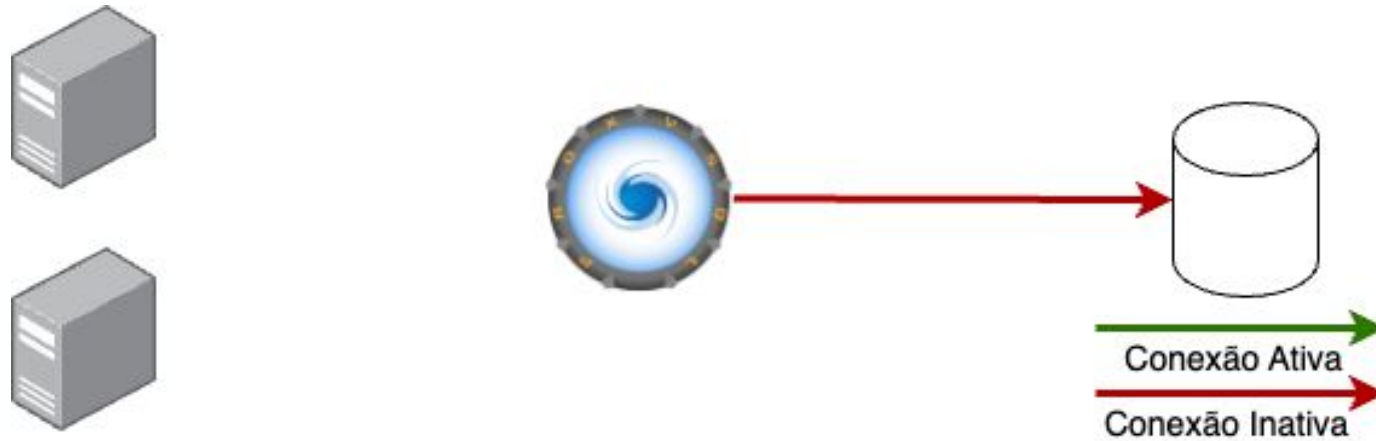
# Features ProxySQL

Connection Pool

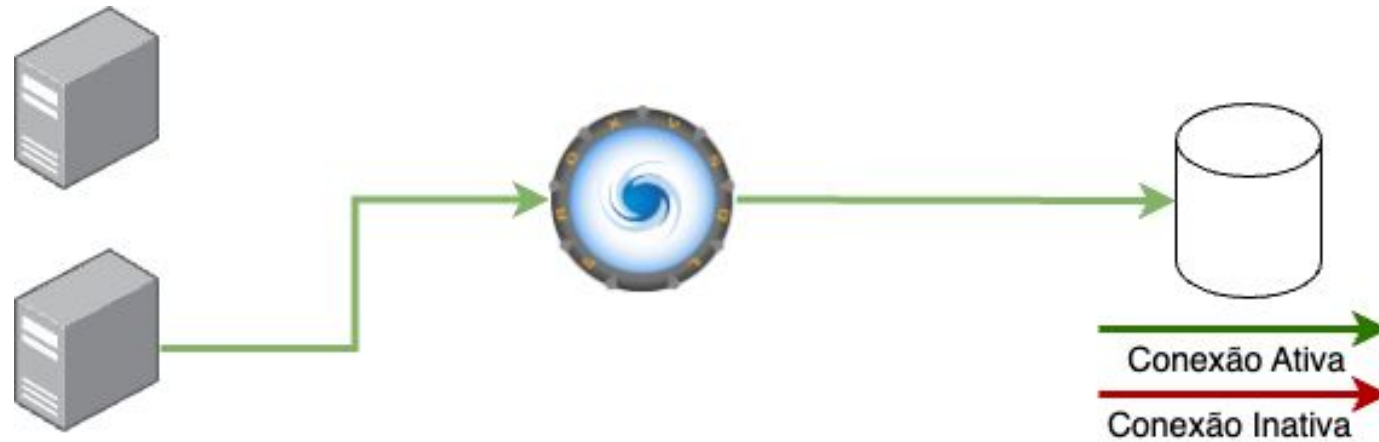
# Connection Pool



# Connection Pool



# Connection Pool

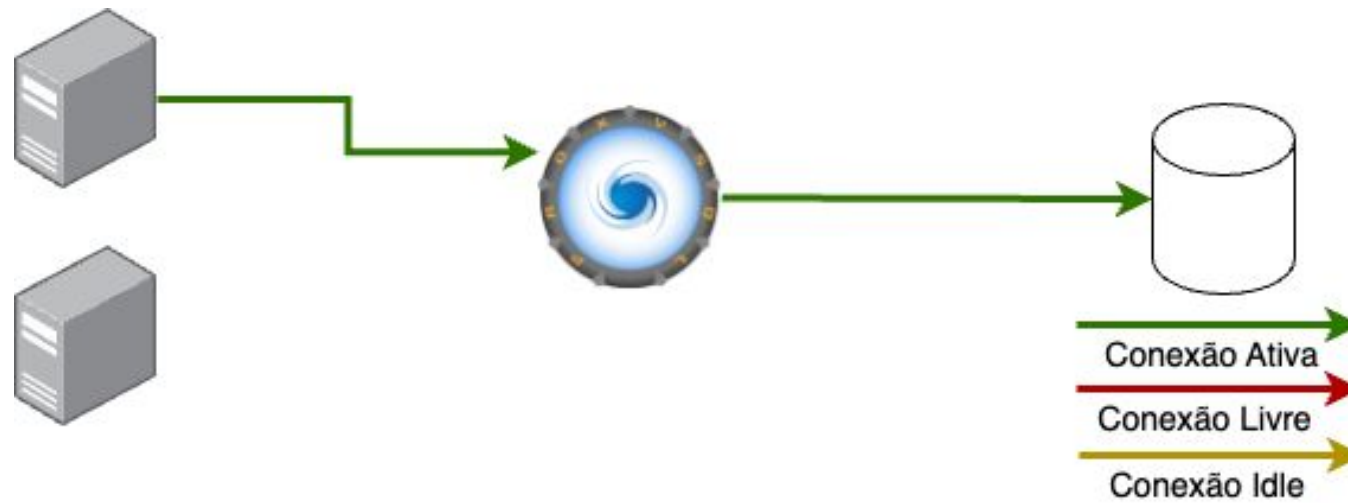




# Features ProxySQL

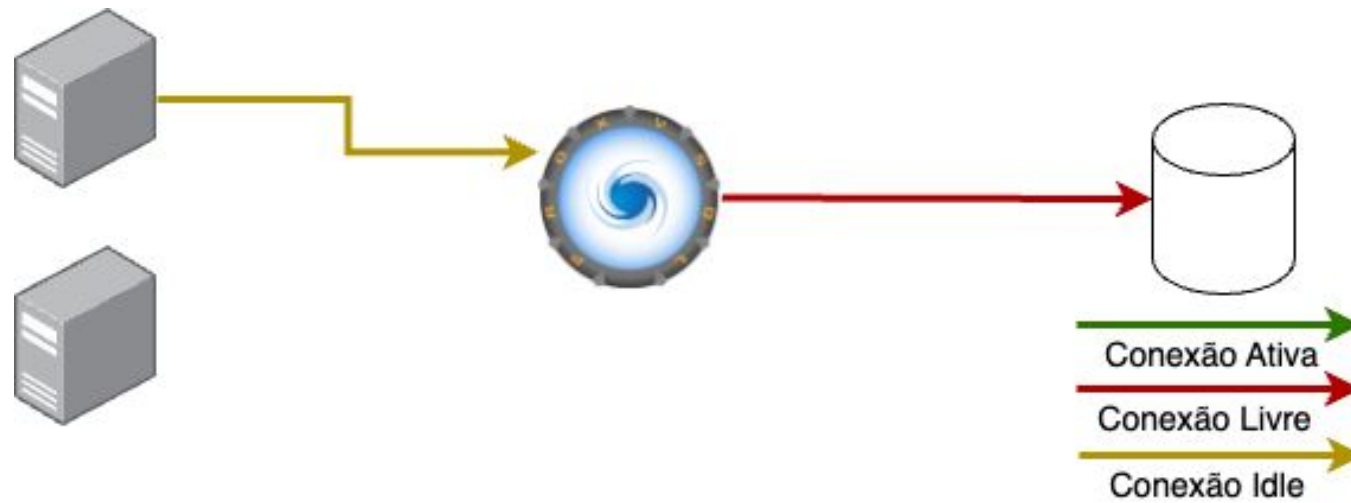
Multiplexing

# Multiplexing

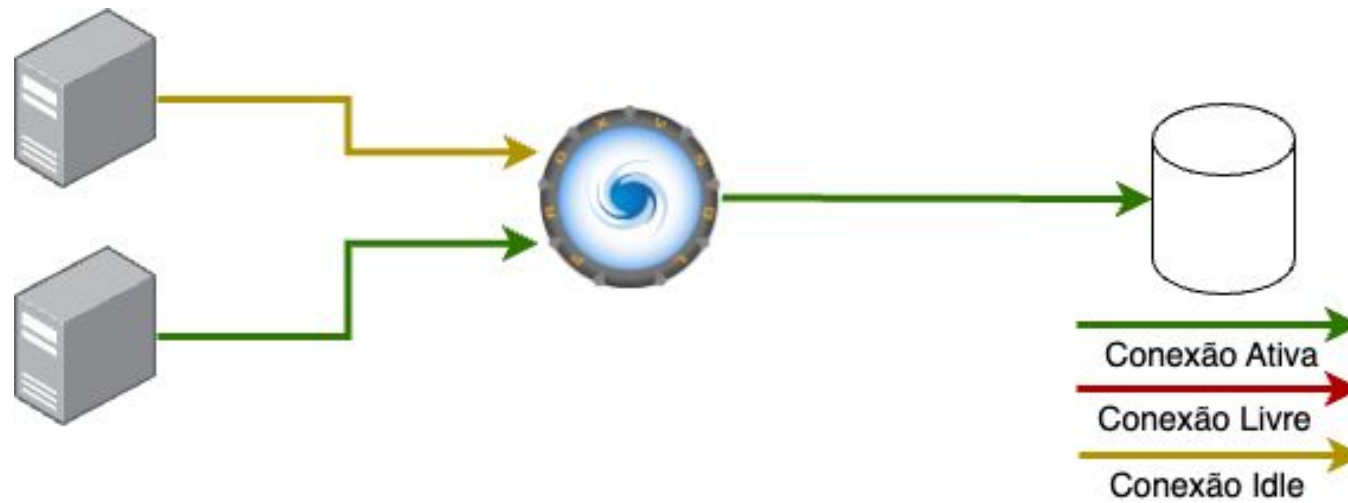




# Multiplexing



# Multiplexing



# Multiplexing – Desabilitar

Condições onde é desabilitado:

- Transação ativa
- Lock em tabela (LOCK TABLE, LOCK TABLES OU FTWRL) – UNLOCK TABLES
- GET\_LOCK()
- SQL\_CALC\_FOUND\_ROWS
- Quando uma tabela temporária é criada (CREATE TEMPORARY TABLE)
- PREPARE
- SQL\_LOG\_BIN=0
- Uso de Variáveis



# Features ProxySQL

Failover

# Failover – Replicação

Variáveis de checagem:

- read\_only
- innodb\_read\_only
- super\_ready\_only
- read\_only|innodb\_read\_only
- read\_only&innodb\_read\_only

```
ProxySQL > SELECT * FROM mysql_replication_hostgroups\G
***** 1. row *****
writer_hostgroup: 1
reader_hostgroup: 2
    check_type: read_only
    comment:
1 row in set (0.00 sec)
```

# Failover – Galera/PXC

Variáveis de check:

- read\_only

```
MySQL [(none)]> select * from mysql_galera_hostgroups\G
***** 1. row *****
      writer_hostgroup: 1
 backup_writer_hostgroup: 3
      reader_hostgroup: 2
 offline_hostgroup: 4
           active: 1
        max_writers: 1
 writer_is_also_reader: 0
max_transactions_behind: 100
           comment: NULL
1 row in set (0.00 sec)
```

# Failover – Group Replication

Variáveis de check:

- read\_only

```
MySQL [(none)]> SELECT * FROM mysql_group_replication_hostgroups\G
***** 1. row *****
      writer_hostgroup: 1
 backup_writer_hostgroup: 3
      reader_hostgroup: 2
 offline_hostgroup: 4
           active: 1
        max_writers: 1
 writer_is_also_reader: 0
max_transactions_behind: 100
           comment:
1 row in set (0.01 sec)
```



# Features ProxySQL

Reescrita de query



# Reescrita de query

## Regra de query utilizando match\_partten

```
[root@roberto-garciadebem-node3 ~]# mysql -uadmin -padmin -h127.0.0.1 -P6032
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 43
Server version: 5.5.30 (ProxySQL Admin Module)
```

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
MySQL [(none)]> INSERT INTO mysql_query_rules (rule_id,active,match_pattern,replace_pattern) VALUES (0,1,'^SELECT \* FROM test\.joinit WHERE i=(.*)$',
'SELECT i,s,t FROM test.joinit WHERE i = \1' );
Query OK, 1 row affected (0.00 sec)
```

```
MySQL [(none)]> LOAD MYSQL QUERY RULES TO RUNTIME;
Query OK, 0 rows affected (0.00 sec)
```

# Reescrita de query

```
[root@roberto-garciadebem-node3 ~]# mysql -uroot -p'verysecretpassword1^' -h127.0.0.1 -P6033 -e'SELECT * FROM test.joinit WHERE i=2'
```

| i | s                                    | t        | g |
|---|--------------------------------------|----------|---|
| 2 | aa3e6d88-8d56-11ee-adec-00163e8ac0a2 | 18:56:25 | 8 |

```
[root@roberto-garciadebem-node3 ~]# mysql -uroot -p'verysecretpassword1^' -h127.0.0.1 -P6033 -e'SELECT * FROM test.joinit WHERE i=2'
```

| i | s                                    | t        |
|---|--------------------------------------|----------|
| 2 | aa3e6d88-8d56-11ee-adec-00163e8ac0a2 | 18:56:25 |



# Features ProxySQL

Bloqueando Queries

# Bloqueando Queries

## Executando query suspeita

```
[root@roberto-garciadebem-node3 ~]# mysql -uroot -p'verysecretpassword1^' -h127.0.0.1 -P6033 -e'SELECT * FROM test.joinit WHERE i=2;
select user,host,authentication_string from mysql.user limit 1'
```

```
+-----+-----+-----+-----+
| i | s | t | g |
+-----+-----+-----+-----+
| 2 | aa3e6d88-8d56-11ee-adec-00163e8ac0a2 | 18:56:25 | 8 |
+-----+-----+-----+-----+
+-----+-----+-----+-----+
| user | host | authentication_string |
+-----+-----+-----+-----+
| repl | % | *A4EAAFACAC7E8241A4C7C903D8E679158895BD32 |
+-----+-----+-----+-----+
```

# Bloqueando Queries

## Verificando Queries

```
MySQL [(none)]> select username,hostgroup,digest_text,digest from stats_mysql_query_digest;
```

| username | hostgroup | digest_text                                                    | digest             |
|----------|-----------|----------------------------------------------------------------|--------------------|
| root     | 2         | SELECT * FROM test.joinit WHERE i=?                            | 0xBB8163FCCB0559B2 |
| root     | 2         | select user,host,authentication_string from mysql.user limit ? | 0x80129DBC0AF9B31  |
| root     | 1         | select @@version_comment limit ?                               | 0x226CD90D52A2BA0B |

3 rows in set (0.00 sec)

# Bloqueando Queries

## Criando regra de bloqueio

```
[root@roberto-garciadebem-node3 ~]# mysql -uadmin -padmin -h127.0.0.1 -P6032
Welcome to the MariaDB monitor.  Commands end with ; or \g.
MySQL [(none)]> INSERT INTO mysql_query_rules (rule_id,active,digest,error_msg) VALUES (0,1,'0x80129DBCD0AF9B31',
'Usuário suspeito/suspeita de SQL Injection');Query OK, 1 row affected (0.00 sec)

MySQL [(none)]> LOAD MYSQL QUERY RULES TO RUNTIME;
Query OK, 0 rows affected (0.00 sec)
```

## Teste

```
[root@roberto-garciadebem-node3 ~]# mysql -uroot -p'verysecretpassword1^' -h127.0.0.1 -P6033 -e'SELECT * FROM test.joinit WHERE i=2;
select user,host,authentication_string from mysql.user limit 1'
+---+-----+-----+-----+
| i | s | t | g |
+---+-----+-----+-----+
| 2 | aa3e6d88-8d56-11ee-adec-00163e8ac0a2 | 18:56:25 | 8 |
+---+-----+-----+-----+
ERROR 1148 (42000) at line 1: Usuário suspeito/suspeita de SQL Injection
```



# Features ProxySQL

Query Cache

# Query Cache

## Query cache baseado em TTL(Time-to-live)

```
MySQL [(none)]> select hostgroup,digest,digest_text,count_star from stats_mysql_query_digest;
```

| hostgroup | digest             | digest_text         | count_star |
|-----------|--------------------|---------------------|------------|
| 2         | 0xF65A07FDA87C64D1 | SELECT * FROM joint | 1          |

```
1 row in set (0.01 sec)
```

```
MySQL [(none)]> INSERT INTO mysql_query_rules (rule_id,active,digest,cache_ttl,apply) VALUES (5,1,'0xF65A07FDA87C64D1',2000,1);
Query OK, 1 row affected (0.00 sec)
```

```
MySQL [(none)]> LOAD MYSQL QUERY RULES TO RUNTIME;SAVE MYSQL QUERY RULES TO DISK;
Query OK, 0 rows affected (0.00 sec)
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
MySQL [(none)]> select hostgroup,digest,digest_text,count_star from stats_mysql_query_digest;
```

| hostgroup | digest             | digest_text         | count_star |
|-----------|--------------------|---------------------|------------|
| -1        | 0xF65A07FDA87C64D1 | SELECT * FROM joint | 1          |
| 2         | 0xF65A07FDA87C64D1 | SELECT * FROM joint | 2          |

```
2 rows in set (0.00 sec)
```





# Features ProxySQL

ProxySQL Cluster

# ProxySQL Cluster – Principais Tabelas

- proxysql\_servers

```
ProxySQL > SELECT * FROM proxysql_servers;
+-----+-----+-----+-----+
| hostname | port | weight | comment |
+-----+-----+-----+-----+
| 10.124.33.77 | 6032 | 0      |          |
| 10.124.33.98 | 6032 | 0      |          |
| 10.124.33.61 | 6032 | 0      |          |
+-----+-----+-----+-----+
3 rows in set (0.01 sec)
```

# ProxySQL Cluster – Principais Tabelas

- Variáveis do Cluster

```
ProxySQL > SELECT * FROM global_variables WHERE variable_name LIKE 'admin-checksum%';
```

| variable_name                    | variable_value |
|----------------------------------|----------------|
| admin-checksum_mysql_query_rules | true           |
| admin-checksum_mysql_servers     | true           |
| admin-checksum_mysql_users       | true           |
| admin-checksum_mysql_variables   | true           |
| admin-checksum_admin_variables   | true           |
| admin-checksum_ldap_variables    | true           |

6 rows in set (0.00 sec)

```
ProxySQL > SELECT * FROM global_variables WHERE variable_name LIKE 'admin-cluster%';
```

| variable_name                                     | variable_value |
|---------------------------------------------------|----------------|
| admin-cluster_username                            | radmin         |
| admin-cluster_password                            | radmin         |
| admin-cluster_check_interval_ms                   | 1000           |
| admin-cluster_check_status_frequency              | 10             |
| admin-cluster_mysql_query_rules_diffs_before_sync | 3              |
| admin-cluster_mysql_servers_diffs_before_sync     | 3              |
| admin-cluster_mysql_users_diffs_before_sync       | 3              |
| admin-cluster_proxysql_servers_diffs_before_sync  | 3              |
| admin-cluster_mysql_variables_diffs_before_sync   | 3              |
| admin-cluster_admin_variables_diffs_before_sync   | 3              |
| admin-cluster_mysql_query_rules_save_to_disk      | true           |
| admin-cluster_mysql_servers_save_to_disk          | true           |
| admin-cluster_mysql_users_save_to_disk            | true           |
| admin-cluster_proxysql_servers_save_to_disk       | true           |
| admin-cluster_mysql_variables_save_to_disk        | true           |
| admin-cluster_admin_variables_save_to_disk        | true           |
| admin-cluster_ldap_variables_diffs_before_sync    | 3              |
| admin-cluster_ldap_variables_save_to_disk         | true           |

18 rows in set (0.00 sec)

# ProxySQL Cluster – Validações

- validações via checksum

```
2023-11-30 19:25:33 [INFO] Cluster: detected a peer 10.124.33.77:6032 with mysql_servers version 4, epoch 1701372331, diff_check 3. Own version: 3, epoch: 1701371936. Proceeding with remote sync
2023-11-30 19:25:33 [INFO] Cluster: detected peer 10.124.33.77:6032 with mysql_servers version 4, epoch 1701372331
2023-11-30 19:25:33 [INFO] Cluster: Fetching MySQL Servers from peer 10.124.33.77:6032 started. Expected checksum 0xFCC21AB57791C8A1
2023-11-30 19:25:33 [INFO] Cluster: Fetching MySQL Servers from peer 10.124.33.77:6032 completed
2023-11-30 19:25:33 [INFO] Cluster: Fetching 'MySQL Group Replication Hostgroups' from peer 10.124.33.77:6032
2023-11-30 19:25:33 [INFO] Cluster: Fetching 'MySQL Galera Hostgroups' from peer 10.124.33.77:6032
2023-11-30 19:25:33 [INFO] Cluster: Fetching 'MySQL Aurora Hostgroups' from peer 10.124.33.77:6032
2023-11-30 19:25:33 [INFO] Cluster: Fetching 'MySQL Hostgroup Attributes' from peer 10.124.33.77:6032
2023-11-30 19:25:33 [INFO] Cluster: Computed checksum for MySQL Servers from peer 10.124.33.77:6032 : 0xFCC21AB57791C8A1
2023-11-30 19:25:33 [INFO] Cluster: Fetching checksum for 'MySQL Servers' from peer 10.124.33.77:6032 successful. Checksum: 0xFCC21AB57791C8A1
2023-11-30 19:25:33 [INFO] Cluster: Writing mysql_servers table
2023-11-30 19:25:33 [INFO] Checksum for table mysql_servers is 0x2700C3A97801C265
2023-11-30 19:25:33 [INFO] Checksum for table mysql_replication_hostgroups is 0x46B740C7F426A83
2023-11-30 19:25:33 [INFO] Rebuilding 'Hostgroup_Manager_Mapping' due to checksums change - mysql_servers { old: 0x12E76C9FDEE03D3A, new: 0x2700C3A97801C265 }, mysql_replication_hostgroups { old:0x46B740C7F426A83, new:0x46B740C7F426A83 }
2023-11-30 19:25:33 [INFO] MySQL_HostGroups_Manager::commit() locked for 2ms
2023-11-30 19:25:33 [INFO] Cluster: Saving to disk MySQL Servers from peer 10.124.33.77:6032
2023-11-30 19:25:34 [INFO] Cluster: detected a new checksum for mysql_servers from peer 10.124.33.61:6032, version 4, epoch 1701372331, checksum 0xFCC21AB57791C8A1 . Not syncing yet ...
2023-11-30 19:25:34 [INFO] Cluster: checksum for mysql_servers from peer 10.124.33.61:6032 matches with local checksum 0xFCC21AB57791C8A1 , we won't sync.
2023-11-30 19:25:34 [INFO] Cluster: checksum for mysql_servers from peer 10.124.33.61:6032 matches with local checksum 0xFCC21AB57791C8A1 , we won't sync.
```

# ProxySQL Cluster – Alterações



A terminal window with two tabs. The active tab is titled 'root@roberto-garciadebem-node3:~ (ssh)' and contains the command: `[root@roberto-garciadebem-node3 ~]# mysql -uadmin -padmin -h127.0.0.1 -P6032 --prompt="ProxySQL > "`. The second tab is titled 'root@roberto-garciadebem-node5:~ (ssh)'.



# Features ProxySQL

Tabelas de estatística

# Stats tables

- SHOW TABLES FROM STATS;
  - Stats\_mysql\_commands\_counters - Estatísticas dos comandos
  - Stats\_mysql\_connection\_pool - Conexões por servidor
  - Stats\_mysql\_global - Estatísticas globais
  - Stats\_mysql\_processlist - SHOW PROCESSLIST
  - Stats\_mysql\_query\_digest - Estatísticas agrupadas por digest
  - Stats\_mysql\_query\_rules - Estatísticas das query rules
  - Stats\_proxysql\_servers\_checksums - Visualização atual do Cluster ProxySQL
    - checksums



# Caso de Uso



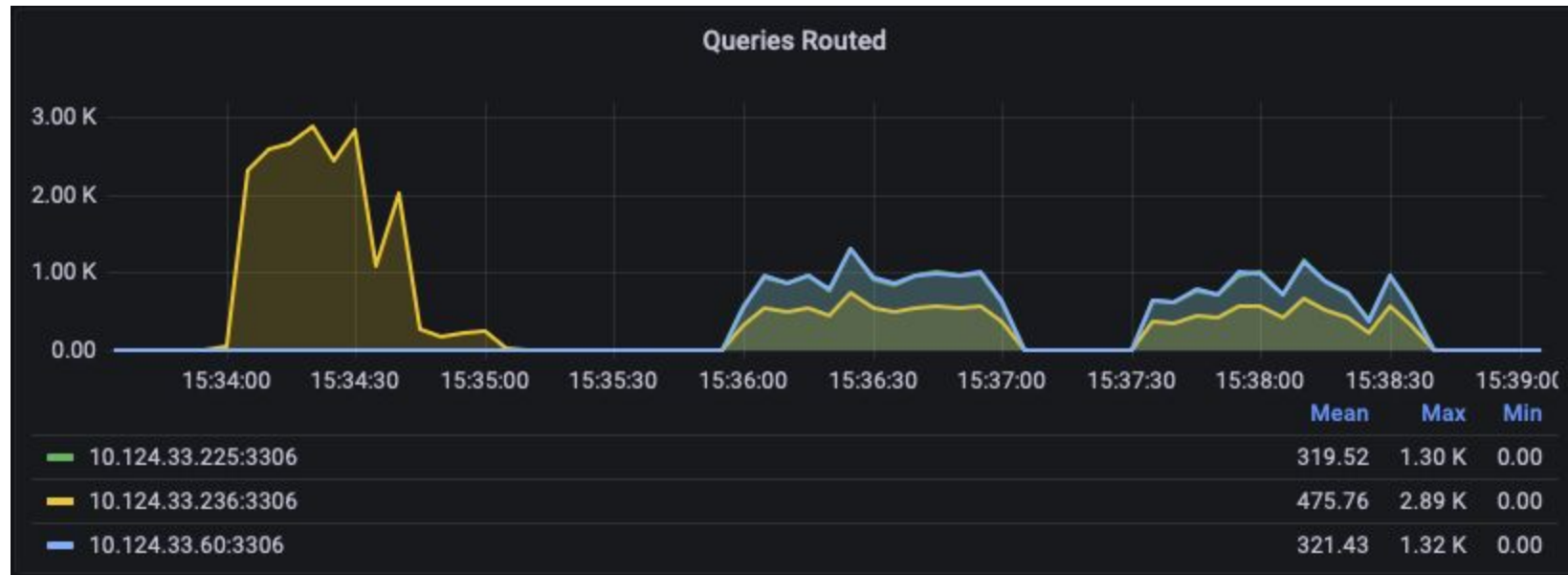
# Caso de Uso

## Sysbench teste

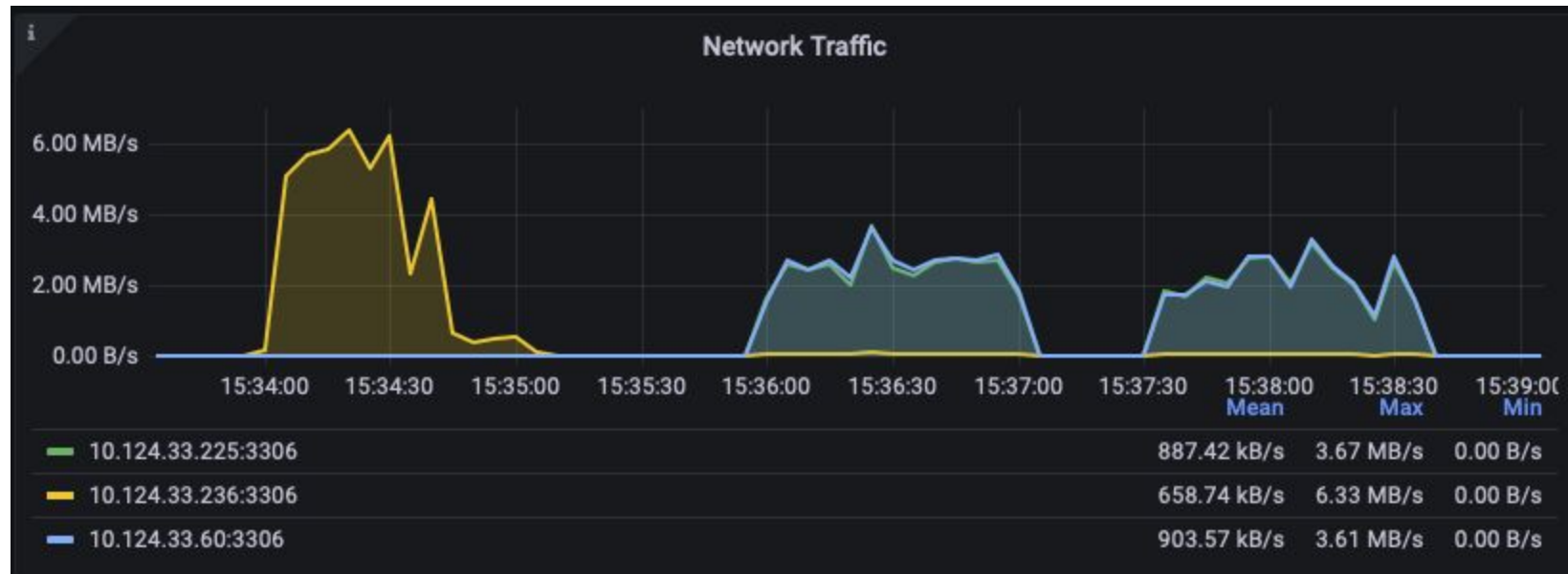
```
sysbench [...] --range_size=100 --table_size=10000 --tables=10  
--threads=20 --events=0 --time=6000 --rand-type=uniform  
/usr/share/sysbench/oltp_read_write.lua
```

- Teste 1 – Todas as conexões no mesmo servidor
- Teste 2 – Fazendo balanceamento de escrita e leitura
- Teste 3 – Fazendo Cache das 10 consultas mais utilizadas

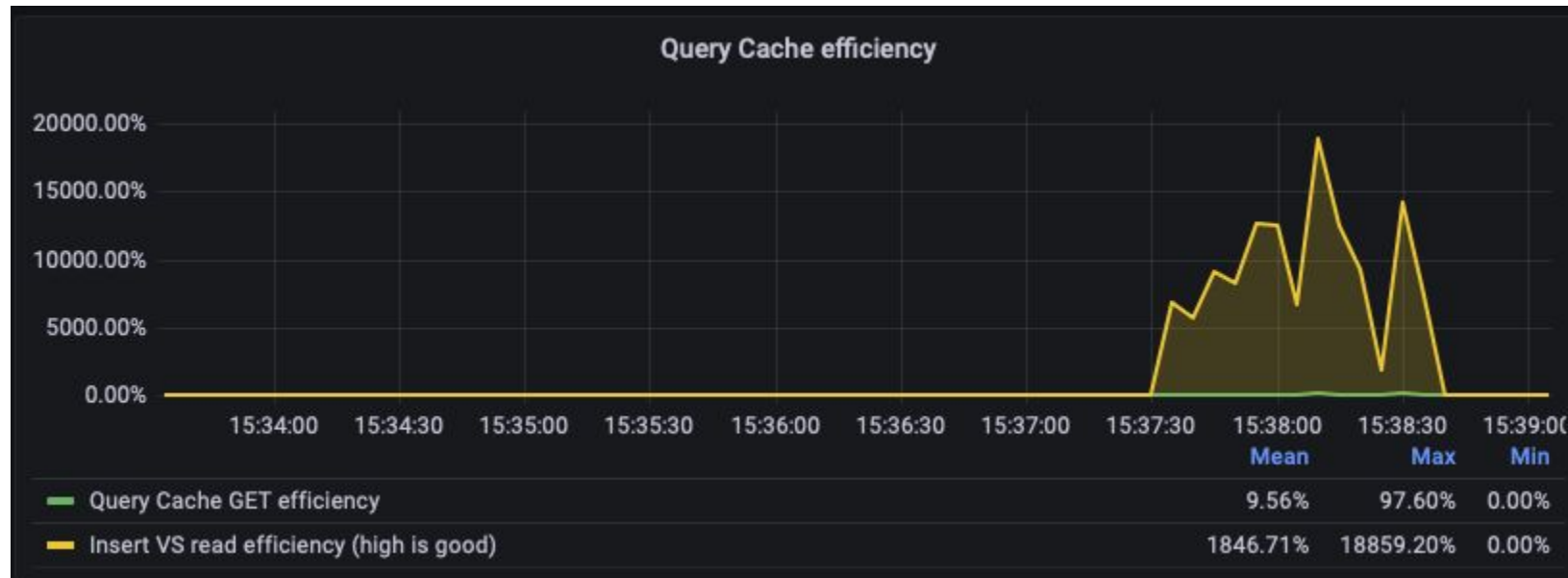
# Caso de Uso



# Caso de Uso



# Caso de Uso



# Perguntas?

Vagas



in





Obrigado!!

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